Harshvardhan Patidar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.67
Sankalp Academy Senior Secondary School, Jhalawar, Rajasthan	Class XII (CBSE)	2024	96.00%
Lady Anusuiya Singhania Education Center, Jhalawar, Rajasthan	Class X (CBSE)	2022	97.40%

SCHOLASTIC ACHIEVEMENTS

- Recipient of the **Academic Excellence Award** for outstanding academic performance among second-year B.Tech. AI students.
- **Inter IIT Tech Meet 14.0:** Part of IIT Hyderabad's team for Adobe Problem Statement; secured **5th** position among all IITs.
- **Codeforces**: Have 1476 rating on Codeforces, demonstrating strong competitive programming skills
- **Meta Hacker Cup 2025**: Cleared Round-1 of the prestigious global programming competition organized by Meta.
- **Secured 3rd Place among 31 teams** in the KLA Hackathon for developing an optimized deep learning-based image denoising solution.

SKILLS

- **Programming Languages :** C, C++, Python
- **AI & Data :** PyTorch, scikit-learn, NumPy, Pandas, Matplotlib
- **Databases :** PostgreSQL, Firebase Firestore, Supabase
- **Tools :** Git, GitHub, Docker, Linux, VS Code, Jupyter Notebook, Google Colab, SSH

RELEVANT COURSES

- **Computer Science:** Algorithms, Data Structures, DBMS - I, Discrete Mathematics
- **AI & Machine Learning:** Foundations of Machine Learning, Deep Learning, Programming for AI, Introduction to Modern AI, Classical AI
- **Mathematics:** Probability & Random Variables, Optimization I & II, Concentration Inequalities, Statistics, Matrix Theory, Numerical Analysis

PROJECTS

Aurora Image Editing Suite | Python, PyTorch, Neural Gaffer, Real-ESRGAN | GitHub

- Engineered a **3-stage** image preprocessing pipeline comprising binary-mask object extraction, square normalization, and HDRI environment map generation to prepare **1024×1024** inputs for Neural Gaffer-based relighting.
- Integrated Neural Gaffer into a remote inference pipeline supporting configurable HDRI, light rotation, and shadow

controls with **6-7 s** inference latency on **A100 GPUs**.
- Integrated Real-ESRGAN into the post-processing pipeline to perform high-quality super-resolution on relit images before final rendering.

Image Denoising Model | Python, PyTorch, NAFNet, DWT | GitHub

- Developed a **NAFNet-based image denoising model** with **Discrete Wavelet Transform (DWT)** to separate noise from structural image details in the frequency domain.
- Optimized deployment using **mixed-precision inference, tile-based processing, and model compilation**, achieving **52 ms inference latency** with a **5.5M-parameter** model.
- Achieved **29.12 dB PSNR** using a hybrid **Charbonnier-FFT loss**, preserving textures and high-frequency details while effectively suppressing noise.

Hybrid File Encryption System | Python, AES-256-CBC, RSA-4096, SHA-256 | GitHub

- Developed a hybrid cryptographic pipeline combining AES-256-CBC for file encryption and RSA-4096 for secure key management.
- Implemented chunk-based processing for large files with SHA-256 checksum verification to ensure data integrity.
- Designed structured JSON metadata for file reconstruction, key storage, and integrity validation.

Adversarial Attacks on ML Fairness | Python, SHAP, Lime | GitHub

- Developed a framework implementing scaffolding and gradient-based adversarial attacks to demonstrate how models can fool LIME/SHAP explainability methods while maintaining demographic parity on real data.
- Built OOD detection and fairness-constrained components (ExponentiatedGradient) that route perturbations intelligently to expose vulnerabilities in post-hoc explainability across Adult Income and COMPAS datasets.
- Implemented CAD-Defend, a defense mechanism filtering adversarial samples to recover true model explanations and validate fairness constraints.

POSITIONS OF RESPONSIBILITY

- **Lambda (Software Development Club), IIT Hyderabad:** Core Member; contributed to the IITH Dashboard by implementing new features and assisting in deployment of application updates.

- **Infero (Competitive Programming Club), IIT Hyderabad:** Core Member; mentored junior students and conducted introductory sessions on data structures, algorithms, and competitive programming.

- **Epoch (AI/ML Club), IIT Hyderabad:** Serving as **Overall Head**, leading the club's technical initiatives, mentorship activities, and student engagement efforts.

- **Epoch (AI/ML Club), IIT Hyderabad:** Served as a **Core Member**, contributing to AI/ML projects and conducting learning sessions for incoming students.

- **Kludge (Cybersecurity and Networking Club), IIT Hyderabad:** Core Member; contributed to club activities and mentoring initiatives for incoming students.